
How Do World Action Models Use Their Futures?

Evidence from Mechanistic Interventions

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Abstract

1 World action models generate future observations alongside robot actions, but co-
2 generation does not show that the future affects the action. We test this directly.
3 Two normal rollouts from the same simulator state produce different future-action
4 pairs, A and B. We then rerun A with its observation, instruction, random draws,
5 and solver fixed, replacing only its future representation with B's. If the new action
6 moves specifically toward action B, the content of the future causally controls
7 action; merely corrupting future features cannot establish this direction. Across
8 22 states and six RoboLab tasks, alternative futures redirect Cosmos 3 actions and
9 executed endpoints almost completely toward the corresponding rollout. Cosmos
10 Policy shows a smaller effect at the first decision state of all ten LIBERO-Long
11 tasks. Matched Gaussian, shuffled, self-recomputation, and alternative natural-
12 future controls rule out perturbation magnitude and generic future replacement
13 as complete explanations. Cosmos Policy responds primarily to camera-visible
14 robot motion, whereas isolated robot or object pixels do not reproduce Cosmos
15 3's whole-future effect. Replacing only future-video attention keys and values
16 removes 83–88% of Cosmos 3 steering in every tested state. These results estab-
17 lish content-specific future conditioning in both policies, while supporting neither
18 candidate comparison nor task-level planning.

19 1 Introduction

20 Many world action models (WAMs) generate future observations alongside robot actions [8, 14].
21 Joint generation, however, does not imply that the action computation uses the generated future.
22 The policy could map the current observation and instruction directly to an action while producing
23 a plausible future as an auxiliary output.

24 The causal test is simple (Figure 1). Starting from one saved state, normal rollout A produces Future
25 A and Action A, while normal rollout B produces Future B and Action B. Their agreement is only
26 correlational: both could be parallel outputs of the same computation. We therefore rerun rollout
27 A while inserting Future B and changing nothing else. The model never receives Action B. We
28 use it only afterward to ask whether the recomputed action moved from Action A toward Action B.
29 Movement toward that particular action is evidence that the specific content of Future B, rather than
30 merely activity at future-token positions, controls action.

31 Future representations could serve many roles: they may be auxiliary outputs, a general computa-
32 tional workspace, a content-specific input such as a prospective robot pose used for inverse dynamics
33 [1, 7], or a representation of consequences used to evaluate candidate actions [2, 3]. These possibil-
34 ities are neither exhaustive nor mutually exclusive. RIFT provides the closest prior causal evidence
35 [22]: masking future reads or editing values at future positions changes trajectories and task success
36 across several WAMs. Such destructive interventions establish dependence on the future channel,

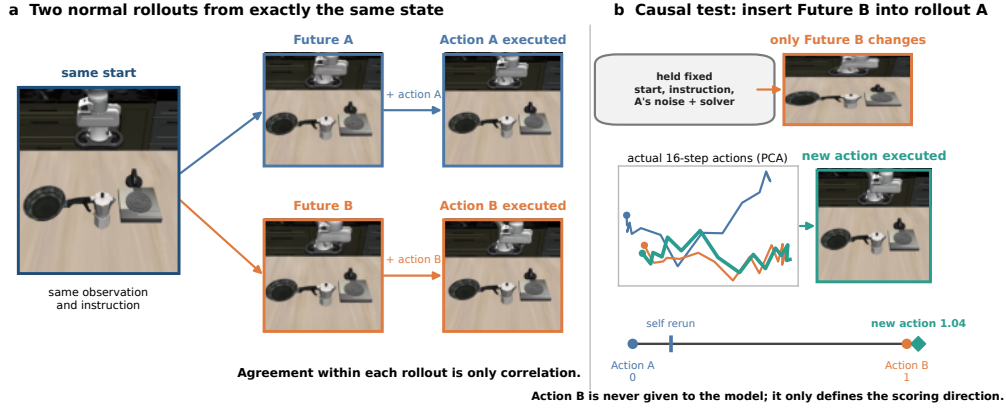


Figure 1: The causal test in plain language, illustrated with an actual held-out LIBERO trace. (a) From the same state, normal rollout A (blue) and rollout B (orange) produce different predicted futures, action chunks, and executed endpoints. This within-rollout agreement is correlational. (b) We rerun A while changing only its future representation to Future B; the observation, instruction, A’s random draws, and solver remain fixed. The 16-step curves are the actual action chunks projected into two dimensions for display. In this example, the recomputed action (green) moves from A toward B; the inference metric uses the full action vector, with A at 0 and B at 1. Action B is withheld from the model and used only for scoring.

but damage can change an action in any direction. They do not show that one coherent alternative future causes the action associated with that future.

We study Cosmos Policy’s direct-policy mode and the diffusion-generator component of Cosmos3-Nano-Policy-DROID. Neither evaluated inference procedure generates and scores multiple candidate actions before acting; Cosmos Policy’s separate best-of- N mode is outside our study. Thus, content-specific future conditioning would not by itself demonstrate outcome comparison or planning.

We ask whether alternative futures steer actions and executed endpoints, which visual content is sufficient, and which internal pathway carries the effect. Cosmos 3 provides the strongest chain: steering generalizes across 22 states, isolated robot or object pixels do not reproduce it, and future-token K/V replacement suppresses action and endpoint steering in every state. Cosmos Policy supplies a cross-generation comparison whose effect is concentrated at early states and is most consistent with camera-visible robot motion.

2 Related Work

Prediction and control. Some visual robot policies generate future frames and infer actions from them; others condition action inference on predicted video representations [1, 7]. Joint video–action models instead learn future observations and controls in one model [6, 8, 10, 14, 20]. Their training objective does not determine whether a particular generated future is used at inference. UVA, for example, can omit video generation at test time [10]. Explicit visual planners answer a different question by sampling alternatives, predicting their outcomes, and comparing them with a goal-dependent score [2, 3].

Recent WAM studies separate predictive training from the use of predictions during action generation. Fast-WAM removes explicit test-time rollout while retaining competitive performance in its evaluations [21]; Faster-WAM retains a lower-cost future-conditioning route and reports improved robustness [24]. AGRA finds that visually plausible foresight need not support reliable action inference in its studied architecture [16]. RIFT directly establishes dependence on future-position values and their organization [22]. Our contribution is complementary: RIFT asks whether behavior depends on the future interface, whereas our intervention asks whether inserting a coherent alternative future predicts the direction of behavior.

66 **Mechanistic interventions.** Activation replacement tests whether an internal representation af-
 67 fects an output rather than merely being decodable from it [12, 18]. Path patching extends this logic
 68 to specified computational routes [5]. Conclusions can depend on the counterfactual pair and met-
 69 ric [23], while multi-component effects need not add when patched and unpatched routes interact
 70 [17]. We therefore treat K/V replacement as evidence for an active pathway, not a complete circuit
 71 decomposition.

72 3 Methods

73 3.1 Causal intervention and outcome

74 Let saved simulator state S admit two unmodified rollouts A and B , with future representations
 75 F_A, F_B , action chunks $\mathbf{a}_A, \mathbf{a}_B$, and endpoints $\mathbf{e}_A, \mathbf{e}_B$ obtained by executing one action chunk. We
 76 recompute A from the identical state while retaining its observation, instruction, action noise, future-
 77 noise path, solver, and denoising schedule, and replace only F_A with F_B . We also test the reciprocal
 78 intervention. For compactness below, we call the recomputed rollout the recipient and the rollout
 79 supplying the alternative future the donor.

80 For an action or endpoint $\mathbf{x} \in \{\mathbf{a}, \mathbf{e}\}$, normalized donor projection is

$$\pi_x(\hat{\mathbf{x}}; A, B) = \frac{(\hat{\mathbf{x}} - \mathbf{x}_A)^\top (\mathbf{x}_B - \mathbf{x}_A)}{\|\mathbf{x}_B - \mathbf{x}_A\|_2^2}. \quad (1)$$

81 The recipient is 0, the donor is 1, and 0.5 is halfway between them along the donor direction; negative
 82 values move away and values above 1 overshoot. Because the dot product retains only this signed
 83 direction, a large change perpendicular to the donor receives little projection. The primary steering
 84 contrast subtracts the corresponding self-future recomputation:

$$\Delta_x = \pi_x(\hat{\mathbf{x}}_{\text{donor}}; A, B) - \pi_x(\hat{\mathbf{x}}_{\text{self}}; A, B). \quad (2)$$

85 The self condition reruns the same intervention machinery with the recipient’s own future, so sub-
 86 traction removes any displacement due merely to recomputation. Pairs are selected using native
 87 action and endpoint separation, never intervention outcomes. Every endpoint arm is executed after
 88 restoring the exact simulator state.

89 3.2 Exact intervention mechanics

90 **Cosmos Policy.** The alternative target is a clean latent encoded with the public tokenizer from
 91 the RGB and proprioceptive endpoint reached by a normal 16-action rollout. In the released latent
 92 layout, temporal indices 5, 6, and 7 are future proprioception, wrist RGB, and primary RGB. At
 93 initialization and at each of the five solver evaluations ($\sigma = 80.0, 42.291, 20.972, 9.618, 4.0$), we
 94 replace only these future coordinates. The denoiser input at noise level σ is $F_B + \sigma \epsilon_A$, where ϵ_A
 95 is the recipient’s fixed future-noise draw, and the returned clean- x_0 estimate is reset to F_B at the
 96 same coordinates. The action and value coordinates are never written. This is a repeated clean- x_0
 97 clamp, not a one-time replacement of a noisy state or predicted velocity. The evaluated direct-policy
 98 configuration is always conditional (its unconditioned input is `None`), so there is no classifier-free-
 99 guidance branch to select; every solver call uses the wrapped x_0 function.

100 **Cosmos 3.** The prepared rectified-flow state is flattened as vision then action: vision has shape
 101 $[1, 48, 9, 33, 40]$ and action $[33, 64]$. We preserve latent frame 0 (the current image) and clamp
 102 future-video frames 1–8. A target F_B is either the final clean video latent from a normal generation
 103 or the public-tokenizer encoding of a 33-frame video produced by executing a normal 32-action
 104 rollout. Before both the conditional and unconditional model forwards at each of the four UniPC
 105 calls, future vision is set to $(1 - \sigma)F_B + \sigma \epsilon_A$. After guidance-3 combines the two predicted velocities,
 106 we replace only future-vision velocity by $(x_\sigma - F_B)/\sigma$, making its clean estimate F_B at the sampler
 107 boundary. The recorded sigmas are 0.999, 0.937, 0.833, 0.624. Thus the clamp is repeated on every
 108 denoising call, occurs after guidance for the output velocity, and never changes the current-video or
 109 action state or action velocity.

110 3.3 Models, populations, and controls

111 **Cosmos Policy.** We evaluate the public `Cosmos-Policy-LIBERO-Predict2-2B` check-
112 point in direct-policy mode on LIBERO-Long, also called LIBERO-10 [8, 11]. The frozen grid
113 contains 30 states: ten tasks crossed with the first query, a state after three open-loop chunks, and
114 a state after six chunks. At each state, eight normal rollouts are generated with fixed seeds. The
115 pair maximizes the mean within-state ranks of action and executed-endpoint distance, subject to
116 prespecified distance floors; both directions and three future-noise draws are then tested. Endpoints
117 concatenate goal-relevant simulator coordinates with the official nine-dimensional proprioceptive
118 endpoint.

119 **Cosmos 3.** We evaluate the 16B `Cosmos3-Nano-Policy-DROID` checkpoint on six eligible
120 RoboLab tasks [14, 15, 19]. Its generator has 36 two-way full-attention layers, 33 raw video frames
121 mapped to nine VAE latent frames, and 32 absolute joint-position actions. Sixteen normal branches
122 are sampled per state; before any intervention, the maximum action-L2 pair is fixed and retained only
123 if its executed endpoints pass the registered separation threshold. A screen using only native suc-
124 cess, separation, and replay validity selected 24 state clusters, capped at four per task; 22 completed.
125 Two source branches ended before producing full 33-frame videos and were left missing rather than
126 replaced. Reciprocal directions and predicted versus executed future sources are repeated measure-
127 ments within state. The natural control is the lowest prespecified seed outside the selected pair;
128 Cosmos Policy natural and shuffled controls are likewise selected without intervention outcomes.

129 Controls include exact self-future recomputation, Gaussian targets matched to the donor in norm and
130 recipient distance, a prespecified natural-future control, within-task shuffled futures, and bitwise
131 no-op checks. For Cosmos Policy, the natural control is the robust same-outcome branch closest
132 to the paired donor in action and endpoint distance when available, and otherwise the preselected
133 distance-matched branch; the shuffled control is the selected alternative from the next registered
134 state of the same task. For Cosmos 3, the natural control is the lowest fixed diffusion seed outside
135 the selected pair. All choices precede intervention outcomes. Gaussian controls test whether an
136 equally large unstructured displacement is sufficient. For Cosmos 3, identity layout capture, sampler
137 wrapping, cache record/replay, and self patches reproduce actions bitwise; sentinels confirm that the
138 transplant does not directly write to action coordinates. Executed-future conditions encode videos
139 produced by physically executing the alternative rollout and replay the same environment prefix
140 before intervention.

141 3.4 Robot and object interventions

142 We isolate content in complementary ways. In Cosmos Policy, first-query interventions replace the
143 wrist-camera future, primary-camera future, or future proprioception separately. A second study
144 crosses binary robot and object states in re-rendered videos at 20 held-out states. A decoder checks
145 whether the intended cell remains visible. A natural-pair screen separately searches policy-generated
146 continuations for robot-divergent/object-matched and object-divergent/robot-matched futures; only
147 four states meet its criteria, below the planned minimum of ten.

148 For Cosmos 3, a separately frozen factor study selected 12 population states round-robin by native
149 object-position separation. Ten completed across all six tasks. Starting from the recipient video, an
150 object-priority disjoint mask copies donor object pixels, donor robot pixels outside the object mask,
151 both, or neither. This 2×2 design tests isolated pixel factors; it does not guarantee an in-distribution
152 video or isolate a latent semantic variable.

153 3.5 K/V interventions

154 In attention, keys help determine what an action query reads, while values carry the information
155 returned by that read. In Cosmos Policy, action queries can attend to future keys. We continuously
156 gate future keys at prespecified block 27, with exact all-key recomputation, equal-count random-key
157 and current-key controls, and a block-0 control. This removal changes attention normalization, so
158 it tests whether a late future-to-action edge is active, not the information carried by a token-count-
159 preserving route.

160 Cosmos 3 permits the stronger replacement test. A self-future run records future-video keys and val-
161 ues separately for all 36 layers and all eight model forwards (four denoising calls, conditional and

unconditional). In the donor-future run, the main pathway test replaces those cached future K/V call by call at every layer and recomputes only the action-query rows. Current-video and future-video query outputs remain native; text, current-video, and action K/V, token count, token order, observation, instruction, noise, and the donor future clamp are unchanged. A separate calibration scan selected layer 16 for single-layer localization, but the reported population pathway estimate uses all 36 direct future-K/V-to-action-query interfaces. We report *K/V replacement loss*: the reduction in donor projection after substituting self-future K/V. This is an operational intervention contrast, not a population natural indirect effect.

3.6 Statistical analysis

The independent unit is a restored simulator state, not an individual noise draw or transplant direction. We first average reciprocal directions, donors, noise draws, and process repeats within state, then form 95% percentile intervals from 10,000 bootstrap resamples of states [4]. For the common-scale condition estimates in Table 1, we reconstruct each control’s projection within unit from the registered pairwise contrasts before bootstrapping; these are not differences of aggregate means. Task-balanced sensitivity analyses average states within task before resampling tasks. We prespecified 0.10 projection units—10% of the native recipient-to-donor displacement—as the smallest effect of interest for the Cosmos Policy confirmatory family and the Cosmos 3 population and factor families [9]. An interval contained within $[-0.10, 0.10]$ supports practical equivalence to zero at that scale. Prespecified primary sign-test p -values use Holm correction.

4 Results

4.1 Do coherent futures steer behavior?

The method succeeds if three things happen: inserting a donor future moves behavior toward that donor, the shift survives simulator execution, and the paired donor outperforms non-donor controls. Table 1 puts the method and every applicable control on one scale, organized by model, future source, and outcome. Figure 2 separates the two statistical questions. Panel a reports donor-minus-self steering, for which positive values indicate movement toward the paired donor and 1 denotes the full native recipient-to-donor displacement. Panel b reports donor-minus-control contrasts, for which positive values mean that the paired donor steers behavior more strongly than the stated control.

Cosmos 3. Across 22 completed states and all six tasks, a predicted paired-donor future produces action steering of 0.992 ($[0.972, 1.010]$; 22/22 states positive), compared with 0.559 for the natural-future control (Table 1). Executed paired-donor videos produce action steering of 0.983, compared with 0.591 for the natural control and 0.041 for the geometry-matched Gaussian control. Executing the paired-donor-conditioned action yields endpoint steering of 1.003 ($[0.950, 1.050]$; 22/22). The corresponding endpoint projections are 0.514 ($[0.387, 0.642]$) for the natural control and -0.041 ($[-0.168, 0.054]$) for the matched Gaussian. In paired secondary analyses, the donor therefore exceeds the natural endpoint control by 0.488 ($[0.348, 0.631]$; 21/22 states positive) and the Gaussian endpoint control by 1.043 ($[0.946, 1.168]$; 22/22). Thus, the result is not explained by recomputation, an equally large unstructured displacement, or merely supplying a different valid continuation. Endpoint projection shows that the donor-directed component survives simulator execution; it does not imply equality of the full endpoint vectors. Task-balanced analyses preserve these conclusions.

Cosmos Policy. At the first query, the paired donor produces action steering of 0.499 and endpoint steering of 0.552, positive in all ten tasks (Table 1). The natural control reaches 0.395 and 0.442, respectively, while the shuffled control reaches 0.244 and 0.275 and the matched Gaussian remains near zero. The natural future is the hardest control: the paired donor exceeds it by 0.103 ($[0.048, 0.161]$; 8/10) for actions and 0.110 ($[0.054, 0.166]$; 8/10) for endpoints. This rules out the weak explanation that any different tokenizer-consistent future causes the same movement. Figure 5 examines how this effect varies across state cohorts and future components.

No screened branch pair in either policy retained a stable success/failure label across continuation seeds. We therefore interpret these results as donor-directed changes in action and one-chunk endpoint, not changes in closed-loop task completion.

Table 1: Method and controls on one scale. Each row reports donor-directed projection after inserting the named future: 0 is the self-future result and 1 is the paired donor behavior. Within each model–source–outcome block, bold marks the largest estimate, always the paired donor. Control projections are reconstructed within independent unit before bootstrapping; the two Cosmos 3 endpoint-control rows are secondary analyses.

Model	Source	Outcome	Future inserted	Projection [95% CI]
Cosmos 3	Predicted	Action	Self	0.000 (reference)
Cosmos 3	Predicted	Action	Natural control	0.559 [0.465, 0.651]
Cosmos 3	Predicted	Action	Paired donor	0.992 [0.972, 1.010]
Cosmos 3	Executed	Action	Self	0.000 (reference)
Cosmos 3	Executed	Action	Matched Gaussian	0.041 [0.003, 0.076]
Cosmos 3	Executed	Action	Natural control	0.591 [0.491, 0.687]
Cosmos 3	Executed	Action	Paired donor	0.983 [0.890, 1.064]
Cosmos 3	Executed	Endpoint	Self	0.000 (reference)
Cosmos 3	Executed	Endpoint	Matched Gaussian	−0.041 [−0.168, 0.054]
Cosmos 3	Executed	Endpoint	Natural control	0.514 [0.387, 0.642]
Cosmos 3	Executed	Endpoint	Paired donor	1.003 [0.950, 1.050]
Cosmos Policy	Predicted	Action	Self	0.000 (reference)
Cosmos Policy	Predicted	Action	Matched Gaussian	0.001 [−0.004, 0.006]
Cosmos Policy	Predicted	Action	Natural control	0.395 [0.269, 0.553]
Cosmos Policy	Predicted	Action	Shuffled control	0.244 [0.152, 0.347]
Cosmos Policy	Predicted	Action	Paired donor	0.499 [0.336, 0.679]
Cosmos Policy	Predicted	Endpoint	Self	0.000 (reference)
Cosmos Policy	Predicted	Endpoint	Matched Gaussian	0.003 [−0.003, 0.008]
Cosmos Policy	Predicted	Endpoint	Natural control	0.442 [0.311, 0.598]
Cosmos Policy	Predicted	Endpoint	Shuffled control	0.275 [0.185, 0.375]
Cosmos Policy	Predicted	Endpoint	Paired donor	0.552 [0.387, 0.728]

213 4.2 What future content causes steering?

214 The content that drives steering differs across checkpoints. Cosmos Policy responds mainly to
215 camera-visible robot motion at early states, whereas neither isolated pixel factor recreates Cosmos
216 3’s whole-future effect.

217 **Cosmos Policy.** Wrist-future action steering is 0.435 ([0.284, 0.605]), compared with 0.101
218 ([0.057, 0.144]) for the primary camera and 0.001 ([−0.004, 0.008]) for future proprioception (Fig-
219 ure 5b). In the 20-state 2×2 study, decoded futures identify the intended cell with 92.5% primary-
220 camera and 98.75% wrist-camera accuracy. The intended object manipulation is therefore visible
221 in the rendered futures, yet the object-state main effect is 0.00015 ([−0.00012, 0.00049]) for ac-
222 tion and 0.00018 ([−0.00040, 0.00102]) for the executed goal-state endpoint. The available analysis
223 does not report the robot-state main effect. Four exploratory natural pairs provide a complementary
224 comparison: robot-motion futures yield action steering of 0.291 ([0.060, 0.521]) and robot-endpoint
225 steering of 0.315 ([0.059, 0.571]), whereas the object-only effects are 0.021 for both action and goal
226 endpoint, with intervals crossing zero. At these sampled early states, the pattern is most consistent
227 with an inverse-dynamics-like role for camera-visible robot motion. It does not show that Cosmos
228 Policy generally ignores object consequences.

229 **Cosmos 3.** Across ten factor-study states and all six tasks, every isolated-factor interval lies inside
230 the prespecified [−0.10, 0.10] equivalence region (Table 2; Figure 3). Panels a and b show all four
231 cells of the 2×2 hybrid intervention: swapping robot pixels, object pixels, or both changes neither
232 the action nor the executed endpoint appreciably. Panel c places the factor effects beside the whole-
233 future intervention on the same axis. Whole-future steering is approximately one unit, whereas robot-
234 pixel effects are 0.0022 for actions and 0.0102 for endpoints, and object-pixel effects are 0.0061 and
235 −0.0001. Neither isolated factor therefore recreates a meaningful fraction of the whole-future effect.
236 The useful signal may be distributed across the image, require robot–object coherence, depend on
237 camera or temporal structure, or interact nonlinearly with masking and tokenization. Hybrid videos
238 may also be off distribution. The experiment establishes insufficiency of each isolated pixel factor,
239 not irrelevance of robot or object information and not joint necessity.

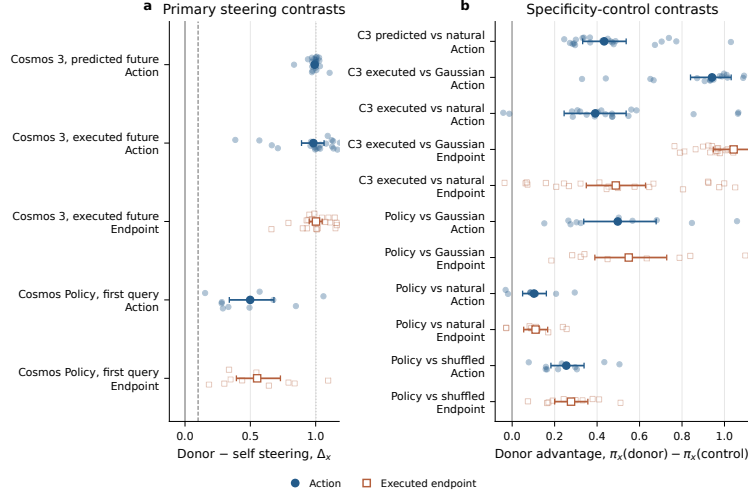


Figure 2: Directional steering and specificity controls. (a) Donor-minus-self steering. Values near 1 mean that the intervention reproduces the donor-directed component of the native action or endpoint displacement; the dashed line marks the prespecified 0.10 smallest effect of interest. (b) Paired donor minus the stated control. Positive values mean that the paired donor has greater directional influence than an unstructured, natural, or shuffled alternative. Small symbols are independent states for Cosmos 3 and tasks for Cosmos Policy; large symbols and whiskers are means and 95% independent-unit bootstrap intervals. Blue circles denote actions and open orange squares denote executed endpoints.

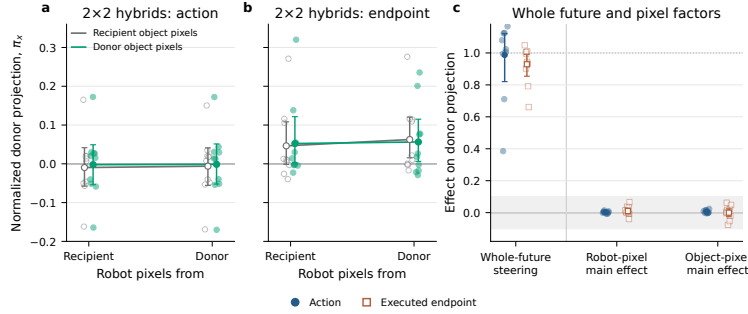


Figure 3: Cosmos 3 robot/object pixel factorization. Panels (a) and (b) show the four hybrid-video cells for action and executed endpoint projection. The horizontal axis indicates whether robot pixels come from the recipient or donor; gray and green traces indicate recipient and donor object pixels. Panel (c) compares the resulting robot- and object-pixel main effects with transplantation of the complete donor future. Blue circles denote actions and open orange squares denote executed endpoints; small symbols are states and large symbols with whiskers are means and 95% state-bootstrap intervals. The gray band is the prespecified $[-0.10, 0.10]$ equivalence region. Isolated robot or object pixels have near-zero effects even though the complete future strongly steers both outcomes.

Table 2: Cosmos 3 isolated-content interventions across 10 states from six tasks. Entries are mean donor-directed projections with 95% state-bootstrap intervals; 0 denotes the recipient and 1 the paired donor. Robot and object masks are disjoint. The prespecified practical-equivalence region is $[-0.10, 0.10]$.

Donor pixels inserted	Action projection [95% CI]	Endpoint projection [95% CI]
Robot only	0.0022 [-0.0016, 0.0060]	0.0102 [-0.0067, 0.0275]
Object only	0.0061 [0.0009, 0.0119]	-0.0001 [-0.0254, 0.0239]

4.3 Which internal pathway carries the effect?

Future-token attention provides an active route from generated futures to action. Population-level K/V replacement in Cosmos 3 gives the stronger of the two pathway tests.

Cosmos Policy. Block-27 future-key removal produces a monotonic action-disruption curve: normalized L_2 change is 0.022, 0.035, 0.054, and 0.068 at gates 0.25, 0.50, 0.75, and 1.00. All-key recomputation is bitwise exact. Full removal is 0.016 more disruptive than equal-count random-key removal and 0.048 more disruptive than block-0 future-key removal; equal-count current-key removal is 0.024 larger. Executed endpoints support the same bounded conclusion, although state counts and intervals are not available for this analysis. The late future-to-action edge is active, but it is neither the only nor the largest route through the block.

Cosmos 3. The interface was selected on excluded calibration tasks and frozen before population testing. Replacing donor-run future-token K/V with self-future K/V removes 0.877 of the 0.992 predicted-action shift, 0.814 of the 0.983 executed-action shift, and 0.851 of the 1.003 endpoint shift (Table 3; Figure 4). These losses correspond to 88%, 83%, and 85% of the respective unpatched point estimates and are positive in 22/22 states. The remaining mean steering after replacement is 0.115, 0.169, and 0.152, respectively (Figure 4a–c). Thus, this interface carries most, but not all, of the donor-directed effect. Task-balanced loss estimates are 0.877, 0.812, and 0.853; all prespecified primary sign tests remain significant after Holm correction. Because token count and attention positions remain fixed, donor-specific information travels through this future-token K/V interface. The experiment does not identify every head or feature, exclude interacting unpatched routes, or establish a complete circuit.

Table 3: Cosmos 3 future-token pathway intervention across 22 states from six tasks. Donor steering is the unpatched donor-directed projection. Replacement loss is the reduction in that projection after replacing donor-future K/V with self-future K/V; intervals are 95% state-bootstrap intervals. Relative reduction is the replacement-loss point estimate divided by unpatched donor steering.

Outcome	Donor steering	Replacement loss [95% CI]	Relative reduction
Predicted action	0.992	0.877 [0.842, 0.910]	88%
Executed action	0.983	0.814 [0.723, 0.900]	83%
Executed endpoint	1.003	0.851 [0.766, 0.922]	85%

4.4 Where is the Cosmos Policy effect concentrated?

Figure 5 summarizes three scope checks for Cosmos Policy. Panel a shows a strong first-query effect, followed by near-zero estimates in the separately sampled states after three and six open-loop chunks: action and endpoint steering are 0.031 and 0.004 after three chunks, and 0.027 and 0.025 after six. This demonstrates state dependence associated with query time, not causal decay within a fixed state. Panel b shows that the early action effect is concentrated in the wrist-camera future (0.435), with a smaller primary-camera effect (0.101) and no measurable proprioceptive effect (0.001). Panel c asks how much more strongly the paired donor steers than each control. Its roughly 0.5 donor-minus-self and donor-minus-Gaussian values reflect the primary effect and the ineffectiveness of matched noise; the smaller natural-control advantage (0.103 for action and 0.110 for endpoint, each positive in 8/10 tasks) is the more stringent content-specific comparison. The shuffled-control advantage is intermediate. Together, these panels localize the Cosmos Policy result to early visual futures while retaining an advantage over other valid futures.

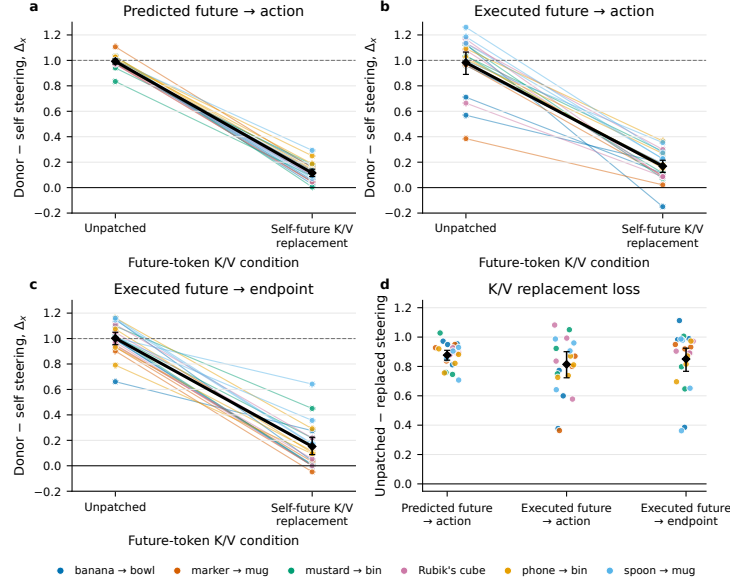


Figure 4: Cosmos 3 future-token K/V replacement. Panels (a)–(c) connect each state’s unpatched donor steering to its value after donor-future keys and values are replaced by self-future keys and values. The black diamonds and whiskers are means and 95% state-bootstrap intervals; colored points identify the six RoboLab tasks. Replacement reduces predicted-action, executed-action, and executed-endpoint steering in all 22 states, leaving mean residual steering of 0.115, 0.169, and 0.152. Panel (d) reports the same intervention as replacement loss (unpatched minus replaced); positive values indicate that the pathway carries donor-specific influence.

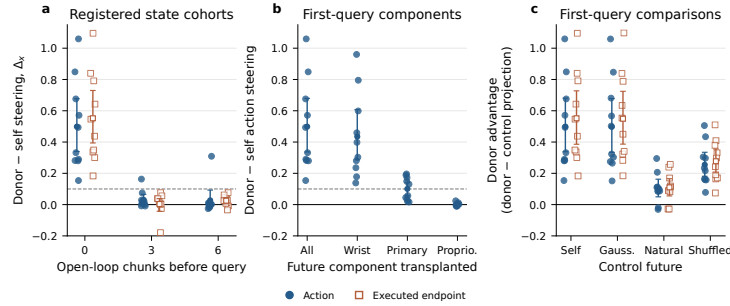


Figure 5: Scope of the Cosmos Policy effect. (a) Paired-donor steering for the first query and for distinct states sampled after three and six open-loop chunks. (b) First-query action steering when all future inputs, only the wrist-camera future, only the primary-camera future, or only future proprioception is transplanted. (c) First-query donor advantage over self, matched-Gaussian, natural-future, and shuffled-future controls. Blue circles denote actions and open orange squares denote executed endpoints; small symbols are tasks and large symbols with whiskers are means and 95% task-bootstrap intervals. The dashed line marks the prespecified 0.10 smallest effect of interest.

274 5 Discussion and Limitations

275 Across both WAM generations, transplanting a coherent future predicts which alternative action
 276 the model will approach. The associated simulator endpoint moves in the same direction. RIFT
 277 established that behavior depends on an organized future interface; our result adds content-specific
 278 sufficiency by showing that a policy-native alternative predicts the signed direction of the change.
 279 The two checkpoints, however, do not appear to rely on the same content.

Cosmos Policy’s early-state pattern resembles an inverse-dynamics interface conditioned on visible robot motion. That account does not transfer cleanly to Cosmos 3: strong whole-future steering coexists with negligible robot-only and object-only pixel effects. Usable content may instead depend on architecture, surrounding context, or distributed feature interactions. The pathway experiments locate where some of this influence travels. Cosmos Policy identifies an active late attention edge, while Cosmos 3 shows across 22 states that future-token K/V carries most of the donor-directed shift.

This evidence is narrower than planning. A planning claim would require evidence that the policy generates alternatives, predicts their outcomes, compares them using a goal, cost, or value, and selects accordingly. We do not test that sequence, and no screened pair retained a stable success/failure label across continuation seeds. Here, “future” and “imagined” describe generated representations, not deliberation. Our claim is that these direct policies condition action on coherent futures, not that WAMs generally plan over task consequences.

Limitations. All interventions are in simulation and cover two checkpoints, six RoboLab tasks, and LIBERO-Long. Cosmos Policy’s strongest result uses one first-query state per task, while later estimates use different physical states and therefore cannot isolate elapsed time. Its natural content analysis contains only four states, and rendered hybrids may be outside the policy’s native future distribution. Cosmos 3 completed 22 of 24 frozen population states and 10 of 12 factor states; the two population omissions resulted from short source videos and were not replaced. Pixel masks may miss distributed semantic features or induce nonlinear changes, so negligible isolated effects do not establish joint necessity. Key removal changes attention normalization, and K/V replacement can interact with unpatched routes. We do not demonstrate task-success mediation or recover a full circuit.

Broader impact. Mechanistic tests can help distinguish plausible-looking predictions from representations that actually control behavior. The same interfaces could also redirect robot actions or create false confidence if validated on narrow state distributions. Use beyond simulation would require broader coverage, physical safety constraints, and monitoring outside the intervention metric.

6 Conclusion

Inserting one normal rollout’s future into another turns “future use” into a directional causal test. Cosmos 3 actions and executed endpoints move almost completely toward the inserted alternative across 22 states and six tasks, and future-video K/V replacement removes 83–88% of that steering. Cosmos Policy shows a smaller early-state effect that survives execution and exceeds natural, shuffled, and geometry-matched controls. These policies use future content to generate actions; whether they compare outcomes to plan remains open.

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A Reproducibility and Protocol Details

A.1 Experimental settings and frozen designs

The Cosmos Policy experiments use the public `Cosmos-Policy-LIBERO-Predict2-2B` checkpoint, `LIBERO-Long`, PyTorch 2.7 with CUDA 12.8, and two NVIDIA H100 80GB GPUs. The frozen design contains ten tasks, three state indices, eight native policy runs per state, one reciprocal donor pair per state, and three future-noise draws per direction. The observation prefix, instruction, recipient action noise, future-noise schedule, tokenizer settings, and denoising schedule are held fixed within an intervention comparison.

The Cosmos 3 experiments use the public `16B Cosmos3-Nano-Policy-DROID` checkpoint with the released Cosmos, Cosmos Framework, and RoboLab evaluation stack. The frozen population screen selected 24 state clusters across six eligible tasks, capped at four clusters per task, using only native success, donor separation, and replay-validity information. Twenty-two clusters completed. The separate robot/object factor study selected 12 clusters by native object-position separation and completed its prespecified minimum of ten across all six tasks.

A.2 State restoration, execution, and missingness

Each endpoint comparison begins from an identical saved simulator state. Evaluation reconstructs the environment, restores that state, replays the observation prefix, verifies the restored observation, and then executes one recomputed action chunk. Recipient and donor endpoints are produced by the same restoration and execution procedure. This design makes the endpoint projection a comparison between reachable continuations of one physical state rather than between unrelated trajectories.

Two frozen Cosmos 3 population units were unavailable because their registered source branches ended after 21 and 29 actions, before producing the required 32-action, 33-frame intervention source. Neither unit was replaced, and no intervention outcome entered the decision to retain or omit a state. The completed population therefore exceeds the prespecified minimum of 20 states across at least five tasks. The factor study likewise meets its minimum of ten states across at least five tasks.

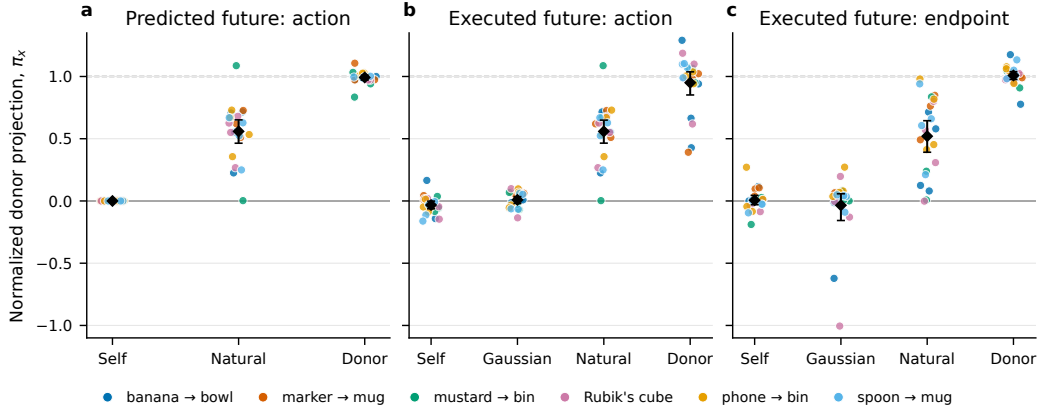


Figure 6: Cosmos 3 condition-level projections across 22 restored states. Panels show predicted-future action, executed-future action, and executed-future endpoint results. Colored points are states grouped by task; black diamonds and whiskers are means and 95% state-bootstrap intervals. Zero denotes the recipient and one the paired donor. Natural controls retain partial donor-aligned content, whereas matched Gaussian targets remain near the recipient and paired donors lie near the native donor.

A.3 Intervention validity and controls

The self-future condition passes the recipient’s own future through the complete intervention machinery. For Cosmos 3, identity token-layout capture, sampler wrapping, cache record/replay, and self-patching reproduce baseline actions bitwise. Sentinels at the sampler boundary confirm that future transplantation does not write directly into action coordinates. The future-token pathway experiment preserves token count and order while leaving current-video, language, action, and all non-targeted key/value states unchanged.

The geometry-matched Gaussian control has the same norm and recipient distance as the donor displacement but no coherent content. Natural controls are valid unpaired continuations selected without intervention outcomes, and shuffled controls use another future from the same task. These comparisons distinguish exact recomputation, perturbation magnitude, generic alternative-future effects, and pair-specific content. Executed-future conditions add a separate realizability check by encoding frames generated when a native donor action is physically replayed in the simulator.

A.4 Statistical conventions

The independent unit is the restored simulator state. Reciprocal directions, donor choices, noise draws, and repeated processes are averaged within state before any population summary. Reported 95% intervals use 10,000 percentile-bootstrap resamples of states; task-balanced checks first average within task and then resample tasks. The 0.10 smallest effect of interest was fixed before the Cosmos Policy confirmatory family and the Cosmos 3 population and factor families were evaluated. Sign tests use state-level directions and Holm correction for each prespecified family.

B Additional Experiments and Diagnostics

B.1 Condition-level Cosmos 3 projections

Figure 6 shows the condition estimates before they are converted into donor-minus-control contrasts. Predicted self-future recomputation stays at zero, natural futures produce partial donor-directed movement, and paired donors cluster near one. For executed futures, the matched Gaussian remains near zero while both action and endpoint effects preserve the same ordering. The state-level display also makes clear that the endpoint conclusion is not produced by a single task or a single extreme unit.

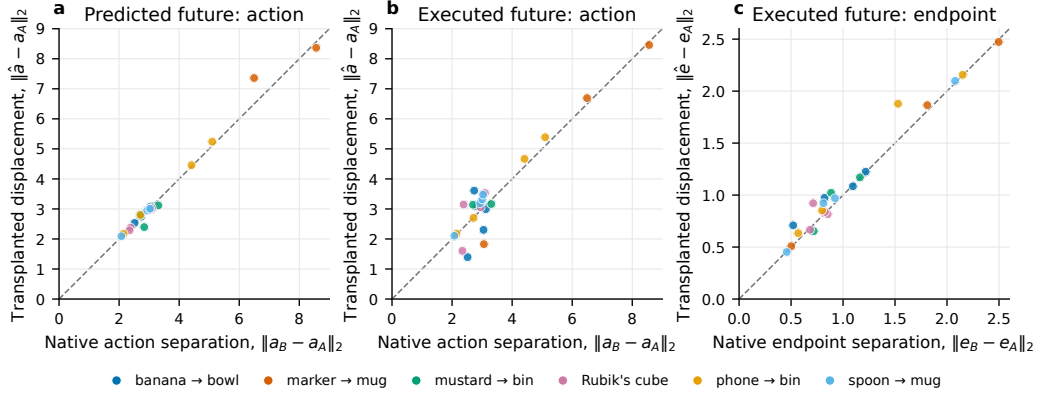


Figure 7: Cosmos 3 magnitude checks. Native recipient-to-donor separation is plotted against the unnormalized displacement caused by donor-future transplantation for predicted actions, executed-future actions, and executed endpoints. The dashed identity line denotes equality of magnitudes; colors identify tasks. These plots complement normalized projection by showing the absolute scale of the induced change.

B.2 Displacement-magnitude diagnostic

Directional projection could be misleading if a large normalized estimate arose from an extremely small native denominator or from a negligible transplanted displacement. Figure 7 therefore compares unnormalized recipient-to-donor separation with the magnitude induced by transplantation. Predicted-action and executed-endpoint displacements lie close to the identity line across the observed range; executed-action displacement is more variable but remains on the scale of the native difference. This diagnostic supports the interpretation that the intervention recreates a substantial behavioral displacement, not merely a tiny change that happens to align with the donor axis.

B.3 Cosmos Policy timing and cross-domain checks

Across all 30 registered Cosmos Policy states, mean donor-minus-self steering is 0.186 ([0.095, 0.290]) for actions and 0.194 ([0.091, 0.305]) for endpoints. The first-query, after-three-chunk, and after-six-chunk action estimates are 0.499, 0.031, and 0.027; the corresponding endpoint estimates are 0.552, 0.004, and 0.025. Because the three indices represent different physical states that may be nested within task trajectories, the first-query analysis is the cleaner confirmatory result and the differences should not be read as causal temporal decay.

The six-state RoboCasa replication is exploratory and is not pooled with LIBERO [13]. Donor-minus-self action steering is 0.0144 ([0.0034, 0.0276]), but its donor-minus-Gaussian contrast is only 0.0025 and has an interval crossing zero. Endpoint steering is 0.0287 ([0.0101, 0.0492]), positive in 6/6 states; donor-minus-Gaussian endpoint steering is 0.0187 ([0.0063, 0.0329]), also positive in 6/6. The replication therefore supports a small executed-state effect while leaving action-space specificity uncertain.

B.4 Within-task relationships in Cosmos Policy

Figure 8 provides two task-level diagnostics at the first query. Action and executed-endpoint steering generally rise together, supporting the use of simulator execution as a behavioral continuation of the action result. Wrist-future steering also tracks all-future steering across tasks, consistent with the aggregate modality decomposition while showing that the wrist view does not reproduce the entire effect uniformly.

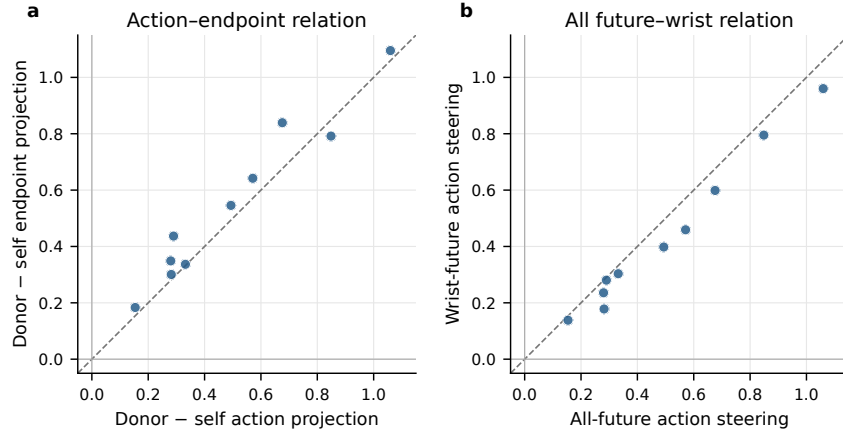


Figure 8: Cosmos Policy task-level relationships at the first query. (a) Donor-minus-self action steering versus executed-endpoint steering. (b) All-future action steering versus wrist-future action steering. Each point is one LIBERO-Long task and the dashed line is identity.

485 B.5 Additional content and pathway checks

486 The Cosmos Policy rendered 2×2 study supplies a manipulation check for the object-content null.
 487 Decoded futures identify the intended hybrid cell with 92.5% accuracy in the primary camera and
 488 98.75% in the wrist camera, while executed endpoints identify the four cells at the 25% chance
 489 rate. Thus, the future representation visibly contains the imposed distinction even though the pre-
 490 registered object main effect on action and goal-state endpoint is approximately zero. The com-
 491plementary natural-pair search found only four states with both an object-divergent/robot-matched
 492and a robot-divergent/object-matched donor at a common recipient; these exploratory data favor the
 493robot-motion donor, but the available population is below the planned minimum of ten.

494 The block-27 Cosmos Policy key-removal analysis includes two additional checks beyond the action-
 495space dose curve reported in the main text. Full future-key removal changes the executed endpoint
 496by an absolute mean projection of 0.281 ([0.164, 0.430]; 10/10 first-query states) and exceeds equal-
 497count random-key removal by 0.169 ([0.058, 0.322]; 9/10). Its difference from equal-count current-
 498key removal is -0.035 with an interval crossing zero. The endpoint analysis therefore supports an
 499active future route at the first query while reinforcing the conclusion that this route is not uniquely
 500dominant over current-state information.

501 For Cosmos 3, task-balanced pathway-loss estimates are 0.877 for predicted actions, 0.812 for
 502executed-future actions, and 0.853 for executed endpoints, closely matching the state-weighted esti-
 503mates. The remaining steering after self-future K/V replacement is 0.115, 0.169, and 0.152, respec-
 504tively. This residual motivates finer-grained tests of parallel attention heads, non-attention routes,
 505and interactions between patched and unpatched components rather than a claim that the selected
 506interface is the complete mechanism.